The postprandial effect of components of the Mediterranean diet on endothelial function.

OBJECTIVES: This study investigated the postprandial effect of components of the Mediterranean diet on endothelial function, which may be an atherogenic factor. BACKGROUND: The Mediterranean diet, containing olive oil, pasta, fruits, vegetables, fish, and wine, is associated with an unexpectedly low rate of cardiovascular events. The Lyon Diet Heart Study found that a Mediterranean diet, which substituted omega-3-fatty-acid-enriched canola oil for the traditionally consumed omega-9 fatty-acid-rich olive oil, reduced cardiovascular events. METHODS: We fed 10 healthy, normolipidemic subjects five meals containing 900 kcal and 50 g fat. Three meals contained different fat sources: olive oil, canola oil, and salmon. Two olive oil meals also contained antioxidant vitamins (C and E) or foods (balsamic vinegar and salad). We measured serum lipoproteins and glucose and brachial artery flow-mediated vasodilation (FMD), an index of endothelial function, before and 3 h after each meal. RESULTS: All five meals significantly raised serum triglycerides, but did not change other lipoproteins or glucose 3 h postprandially. The olive oil meal reduced FMD 31% (14.3 +/- 4.2% to 9.9 +/- 4.5%, $p = 0.008$). An inverse correlation was observed between postprandial changes in serum triglycerides and FMD ($r = -0.47$, $p < 0.05$). The remaining four meals did not significantly reduce FMD. CONCLUSIONS: In terms of their postprandial effect on endothelial function, the beneficial components of the Mediterranean and Lyon Diet Heart Study diets appear to be antioxidant-rich foods, including vegetables, fruits, and their derivatives such as vinegar, and omega-3-rich fish and canola oils.